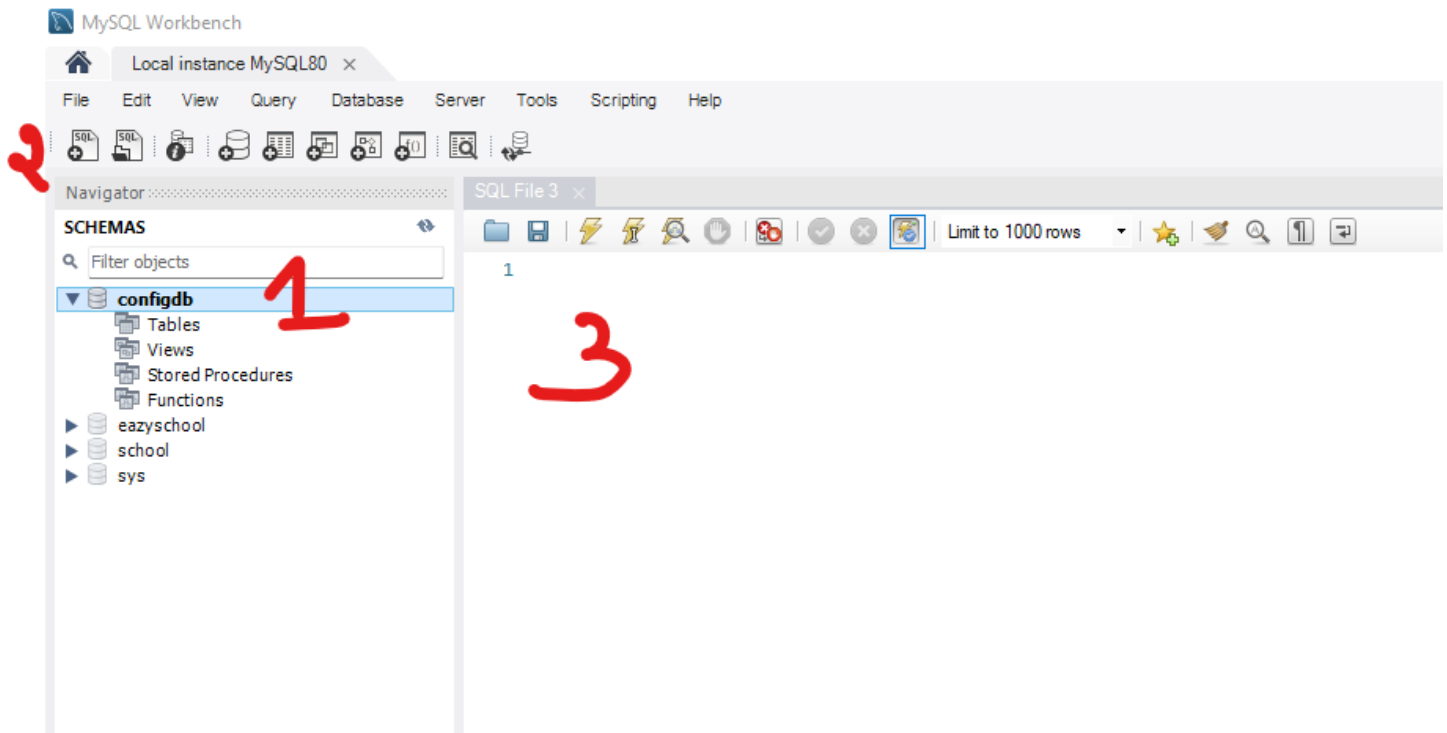
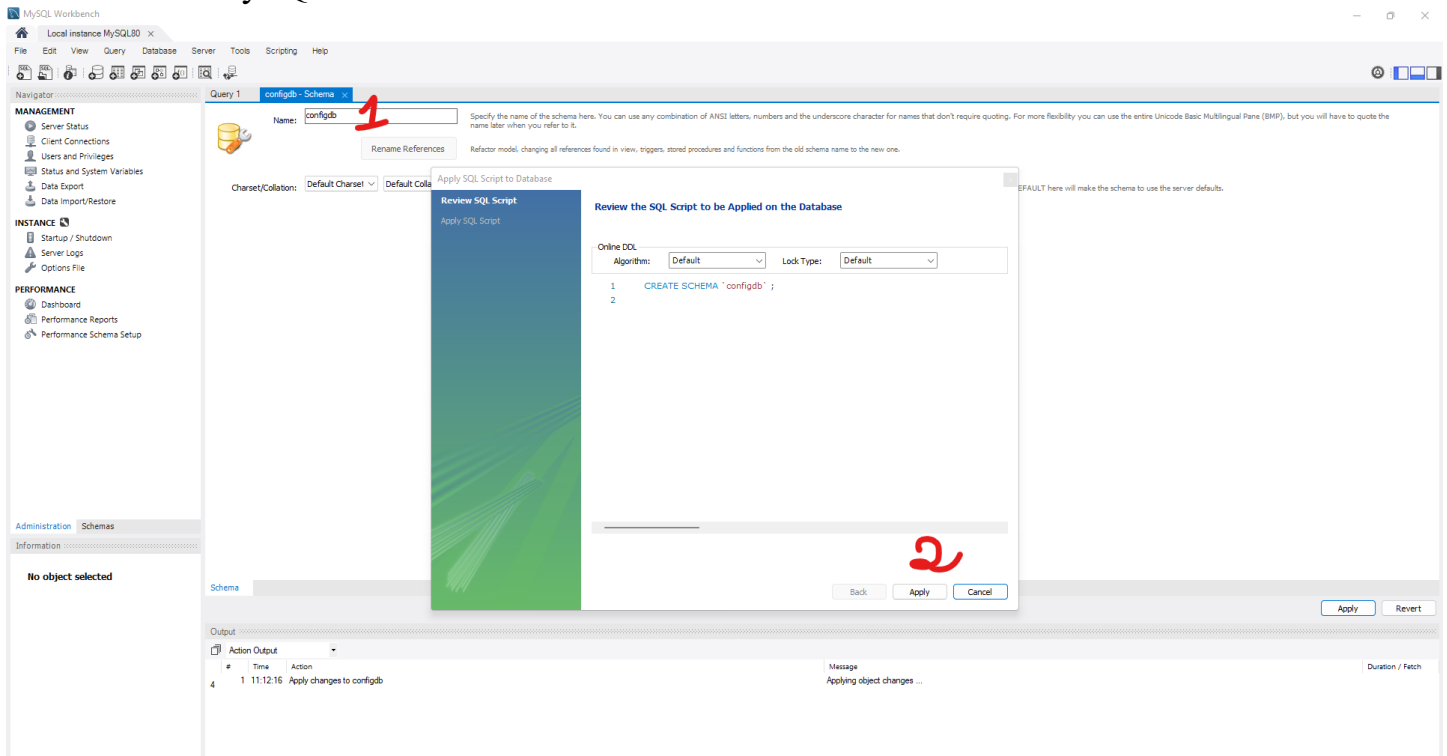


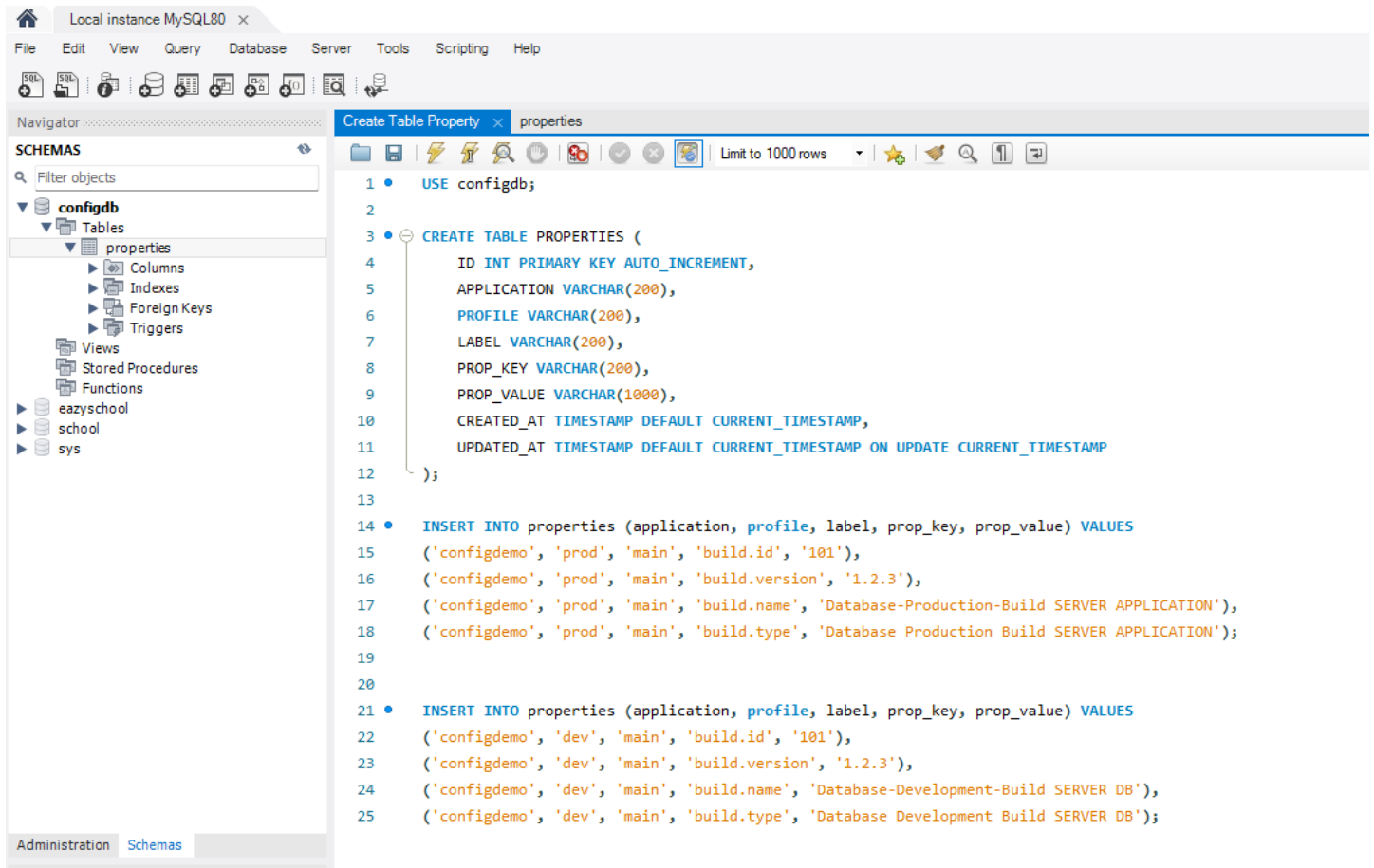
# Using Database to store configuration

## 1- Create MySQL database



- Is the connection not working , check the cmd+r : services.msc

| Name                      | Description                                                                                                                       | Status  |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------|
| MySQL80                   |                                                                                                                                   | Running |
| Windows Defender Firewall | Windows Defender Firewall helps protect your computer by preventing unauthorized users from gaining access to your computer th... | Running |



USE configdb;

```
CREATE TABLE PROPERTIES (
  ID INT PRIMARY KEY AUTO_INCREMENT,
  APPLICATION VARCHAR(200),
  PROFILE VARCHAR(200),
  LABEL VARCHAR(200),
  PROP_KEY VARCHAR(200),
  PROP_VALUE VARCHAR(1000),
  CREATED_AT TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  UPDATED_AT TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

```
INSERT INTO properties (application, profile, label, prop_key, prop_value) VALUES
('configdemo', 'prod', 'main', 'build.id', '101'),
('configdemo', 'prod', 'main', 'build.version', '1.2.3'),
('configdemo', 'prod', 'main', 'build.name', 'Database-Production-Build SERVER APPLICATION'),
('configdemo', 'prod', 'main', 'build.type', 'Database Production Build SERVER APPLICATION');
```

```
INSERT INTO properties (application, profile, label, prop_key, prop_value) VALUES
('configdemo', 'dev', 'main', 'build.id', '101'),
('configdemo', 'dev', 'main', 'build.version', '1.2.3'),
('configdemo', 'dev', 'main', 'build.name', 'Database-Development-Build SERVER DB'),
('configdemo', 'dev', 'main', 'build.type', 'Database Development Build SERVER DB');
```

```
SELECT PROP_KEY, PROP_VALUE FROM PROPERTIES WHERE APPLICATION = 'configdemo' AND PROFILE =
'dev' AND LABEL = 'main';
```

| Result Grid |      |             |         |       |               |                                              |                     |                     |
|-------------|------|-------------|---------|-------|---------------|----------------------------------------------|---------------------|---------------------|
|             | ID   | APPLICATION | PROFILE | LABEL | PROP_KEY      | PROP_VALUE                                   | CREATED_AT          | UPDATED_AT          |
| ▶           | 1    | configdemo  | prod    | main  | build.id      | 101                                          | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 2    | configdemo  | prod    | main  | build.version | 1.2.3                                        | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 3    | configdemo  | prod    | main  | build.name    | Database-Production-Build SERVER APPLICATION | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 4    | configdemo  | prod    | main  | build.type    | Database Production Build SERVER APPLICATION | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 5    | configdemo  | dev     | main  | build.id      | 101                                          | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 6    | configdemo  | dev     | main  | build.version | 1.2.3                                        | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 7    | configdemo  | dev     | main  | build.name    | Database-Development-Build SERVER DB         | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
|             | 8    | configdemo  | dev     | main  | build.type    | Database Development Build SERVER DB         | 2026-06-16 11:21:06 | 2026-06-16 11:21:06 |
| *           | NULL | NULL        | NULL    | NULL  | NULL          | NULL                                         | NULL                | NULL                |

## 2- Add dependency in the config-server



**Project**

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ **Java** ☐ Kotlin ☐ Groovy

☒ **Maven**

**Spring Boot**

☐ 4.1.1 (SNAPSHOT) ☒ **4.1.0** ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7 ☐ 3.5.15

**Project Metadata**

Group

Artifact

Package name

Packaging ☒ **Jar** ☐ War

Configuration ☐ Properties ☒ **YAML**

Java ☐ 26 ☒ **25** ☐ 21 ☐ 17

### Dependencies

ADD DEPENDENCIES... CTRL + B

#### JDBC API SQL

Database Connectivity API that defines how a client may connect and query a database.

#### MySQL Driver SQL

MySQL JDBC driver.

GENERATE CTRL + G

EXPLORE CTRL + SPACE

...

```

1 server:
2   port: 8888
3 spring:
4   application:
5     name: configserver
6 # 1. Activate the jdbc profile here
7   profiles:
8     active: jdbc
9   datasource:
10    url: jdbc:mysql://localhost:3306/configdb
11    username: root
12    password: admin
13    driver-class-name: com.mysql.cj.jdbc.Driver
14 cloud:
15   config:
16     server:
17       jdbc:
18         sql: "SELECT PROP_KEY, PROP_VALUE FROM PROPERTIES WHERE APPLICATION = ? AND PROFILE = ? AND LABEL =?"
19         default-label: main

```

```
server:
  port: 8888
spring:
  application:
    name: configserver
# 1. Activate the jdbc profile here
profiles:
  active: jdbc
datasource:
  url: jdbc:mysql://localhost:3306/configdb
  username: root
  password: admin
  driver-class-name: com.mysql.cj.jdbc.Driver
cloud:
  config:
    server:
      jdbc:
        sql: "SELECT PROP_KEY, PROP_VALUE FROM PROPERTIES WHERE APPLICATION = ? AND PROFILE = ?
AND LABEL =?"
        default-label: main
```

- Importance: we need to # 1. Activate the jdbc profile by default it will get from git hub.

## 1. What does `jdbc` mean?

**JDBC stands for Java Database Connectivity.**

It is a standard Java technology that allows a Java application to talk to a database (like MySQL, PostgreSQL, or Oracle).

- o Without JDBC, your Java code cannot send SQL queries or save data to MySQL.
- o In your project, you added the `mysql-connector-j` dependency. That connector acts as the translator between Spring Boot and your MySQL database using JDBC rules.

## 2. What does `profiles: active: jdbc` mean?

In Spring Boot, a **Profile** is like a "mode" or a "configuration switch." It allows you to say, "When this switch is on, activate this specific set of features or settings."

By setting `profiles.active: jdbc` inside your **Config Server**, you are telling it:

**"Hey Config Server, I want you to run in Database Mode instead of your default Git Mode."**

## How Spring Cloud Config Uses This Switch

Spring Cloud Config Server is built to support many different places to store configuration data (Git, a Database, Vault, or Local files).

### 3-Client config

- Add dependencies for client

```
<!-- Spring Cloud Config Client -->
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-config</artifactId>
</dependency>
```

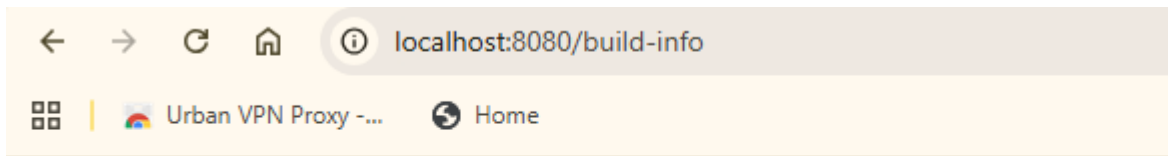
- Application.yml

```
1 spring:
2   application:
3     name: configdemo
4   profiles:
5     active: dev
6   config:
7     import: optional:configserver:http://localhost:8888
8
9 build:
10  id: ${ID}
11  version: ${VERSION}
12  name: ${NAME}
13
14 management:
15  endpoints:
16    web:
17      exposure:
18        include: refresh
```

```
spring:
  application:
    name: configdemo
  profiles:
    active: dev
  config:
    import: optional:configserver:http://localhost:8888
```

```
build:
  id: ${ID}
  version: ${VERSION}
  name: ${NAME}
```

```
management:
  endpoints:
    web:
      exposure:
        include: refresh
```



Build ID: 101, Version: 1.2.3, Name: Database-Development-Build SERVER DB

## - Setup controller

```
BuildInfo.java X
config-demo > src/main/java > com.info.config > BuildInfo > name: String
1 package com.info.config;
2
3+ import org.springframework.boot.context.properties.ConfigurationProperties;
7
8 @Configuration
9 @ConfigurationProperties(prefix = "build")
10 @Data
11 public class BuildInfo {
12
13     private String id;
14     private String version;
15     private String name;
16
17 }
```

```
BuildInfo.java  ConfigApplication.java  BuildInfoController.java X
config-demo > src/main/java > com.info.config > BuildInfoController >
1 package com.info.config;
2
3+ import org.springframework.beans.factory.annotation.Value;
7
8 @RestController
9 // @AllArgsConstructor
10 @RefreshScope
11 public class BuildInfoController {
12
13     @Value("${build.id:default}")
14     private String buildId;
15
16     @Value("${build.version:default}")
17     private String buildVersion;
18
19     @Value("${build.name:default}")
20     private String buildName;
21
22     @GetMapping("/build-info")
23     public String getBuildInfo() {
24         return "Build ID: " + buildId + ", Version: " + buildVersion + ", Name: " + buildName;
25     }
26 }
```

# យើងមើលកន្លែង

config:

import: optional:configserver:http://localhost:8888

ជាកន្លែងដែលយើងធ្វើការភ្ជាប់ទៅកាន់ config server

- Log file

```
2026-06-16T13:44:53.073+07:00 INFO 27176 --- [configdemo] [ restartedMain] com.info.config.ConfigApplication : Starting ConfigApplication using Java 25.0.1 with
PID 27176 (C:\Users\home\Desktop\UdemyCourse\udemy-microservice\softcode\config\target\classes started by home in C:\Users\home\Desktop\UdemyCourse\udemy-microservice
\softcode\config)
2026-06-16T13:44:53.073+07:00 INFO 27176 --- [configdemo] [ restartedMain] com.info.config.ConfigApplication : The following 1 profile is active: "dev"
2026-06-16T13:44:53.075+07:00 INFO 27176 --- [configdemo] [ restartedMain] o.s.c.c.c.ConfigServerConfigDataLoader : Fetching config from server at :
http://localhost:8888
2026-06-16T13:44:53.075+07:00 INFO 27176 --- [configdemo] [ restartedMain] o.s.c.c.c.ConfigServerConfigDataLoader : Located environment: name=configdemo, profiles=
[default], label=null, version=null, state=null
2026-06-16T13:44:53.076+07:00 INFO 27176 --- [configdemo] [ restartedMain] o.s.c.c.c.ConfigServerConfigDataLoader : Fetching config from server at :
http://localhost:8888
2026-06-16T13:44:53.076+07:00 INFO 27176 --- [configdemo] [ restartedMain] o.s.c.c.c.ConfigServerConfigDataLoader : Located environment: name=configdemo, profiles=
[dev], label=null, version=null, state=null
2026-06-16T13:44:53.322+07:00 INFO 27176 --- [configdemo] [ restartedMain] o.s.cloud.context.scope.GenericScope : BeanFactory id=36c9816c-74a3-30ab-a579-
32257a9038a2
```